Artificial Intelligence (AI): Refers to the simulation of human intelligence in machines, enabling them to perform tasks that typically require human cognition:

- Problem solving
- Decision making
- Understanding language
- Recognising patterns

Machine Learning (ML): A subset of AI that involves training machines to learn from data and make predictions or decisions without being explicitly programmed. ML algorithms analyse patterns in historical data and use them to predict future outcomes

- ML is the process of training a software program, called a model, to analyze a dataset and make useful predictions
- Once trained, the mode can generate predictions for new, unseen data. These predictions help automate decision-making

Deep Learning (DL): A specialized branch of ML that uses neural networks with multiple layers (DNN) to model complex patterns in data. DL mimics how the human brain processes information, enabling machines to perform tasks

Steps:

1. Data Collect
2. Data Process
3. Data Preprocess
4. Feature Engineer
5. Training
6. Validate
7. Hyperparameter tuning
8. Deploy

Rule-based Human involved in all aspects		**Machine learning** No human involved
Deterministic	**System characteristics**	Probabilistic
None	**Learning algorithm**	Statistical rules are automatically learned by machines
Not scalable	**Project scale**	Can be easily scaled
Small and simple data	**Dataset**	Large and good quality data for accurate prediction

Use Rule-based system:

- The situations in which there are lower volumes of data
- The rules are relatively simple
- Require speedy output
- When there is a clear and constant data distribution

Use ML:

- When simple rules do not apply
- To handle complex rules and intensive issues with a relatively variable environment
- For long terms as it is more manageable to constant improvement and enhancement through algorithm and data preparation
- Data distinction changes over time, requiring constant adaptation of programs

Supervised Learning:

- Regression:
  - Regression predictive modeling is the task of approximating a mapping function (F) from input variables (X) to a **continuous output variable (Y)**
  - A continuous output variable is a real-value (integer or float). These are often quantities, such as amounts and sizes
- Classification:
  - Classification predictive modeling is the task of approximating a mapping function (F) from input variables (X) to **discrete output variables (Y)**
  - The output variables are often called labels or categories. The mapping function predicts the class or category for a given observation

DL is a class of techniques in ML, it consists of multiple processing layers that allow representation learning of multiple level data abstraction

- Representation Learning:
  Input → Features learning → Mapping from features → Output
- Deep Learning:
  Input → Simplest features → Most complex features → Mapping from features → Output

2 types of DL:

- Convolutional Neural Network (CNN):
  - Image classification
  - Convolve: Roll up to higher levels of abstraction in feature sets
- Recurrent Neural Network (RNN):
  - Speech, text, audio recognition
  - Recur: Iterate over sequential inputs with a memory function

Neural Network's mechanism:

- Structural system based on cascading layers of neurons with variable parameters: weight and bias
- Vary the weights and biases to see if a better outcome is obtained
- Repeat until the network correctly classifies the data
- Backpropagation: It first calculates the total error L(w) then the contribution to the error at each step going backwards

Machine Learning Life Cycle: A sequence of steps followed to develop a machine learning model. These steps ensure that the model is designed, developed and deployed effectively

Industry-recognised Life Cycles:

- SEMMA (Sample, Explore, Modify, Model, Assess): Developed by SAS, this framework emphasises the modeling aspect
- TDSP (Team Data Science Process): Microsoft's agile method for enhancing team collaboration in data science
- KDD (Knowledge Discovery in Databases): An older framework for extracting knowledge from databases
- CRISP-DM (Cross-Industry Standard Process for Data Mining): A standard data mining process also used in machine learning
    - Business Understanding: Identify problems, define goals and establish success metrics
    - Data Understanding: Collect, explore and assess data quality
    - Data Preparation: Clean, construct, integrate and format data for analysis
    - Modeling: Select techniques, build models, adjust parameters
    - Evaluation: Assess the model's performance, ensure alignment with business objectives
    - Deployment: Integrate models into production for use

Deep Learning Frameworks:

- A DL framework is an interface, library or a tool that allows us to build DL models more easily and quickly without delving into the details of underlying algorithms
- Developers no longer need to compile code from scratch using complex neural networks and backpropagation algorithms. Instead, they can use existing models to configure parameters as required, where the model parameters are automatically trained
- Moreover, they can add custom network layers to the existing models or directly select required classifiers and optimization algorithms by invoking existing code

TensorFlow:

- Reasons to choose:
  - Scalability: Scale the model from a single GPU to multiple GPUs and even across multiple machines using simple APIs that require minimal changes to the code
  - GPU Support: Run computations on various types of devices (CPUs or GPUs)
  - Multi-lingual: Support many languages, currently Python is the best supported
  - Multi-platform: Available on 64-bit Linux and Windows as well as Android and IOS
  - Powerful Computing
  - Distributed: Run on many devices, from smartphones to large-scale computer clusters
- Model compilation: Involves 3 parts:
  - Loss function: Measures how accurate the output is
  - Optimizer: Measures how the model is updated
  - Metrics: Monitors the training to determine when to stop
  - Example:

```
model.compile(optimizer=tf.train.AdamOptimizer(),
              loss=tf.keras.losses.categorical_crossentropy,
              metrics=[tf.keras.metrics.categorical_accuracy])
```

- Model Training:
  - All data is trained through batch iteration, we use model.fit in TensorFlow
  - An Epoch refers to 1 complete pass through the entire training dataset
- Model Evaluate:
  - In TensorFlow, we use model.evaluate for testing
  - It compares the predicted results with the groundtruth label to calculate the accuracy of the test set

Areas of ML:

- Computer Vision:
  - Image Recognition: Identify objects, people, text or scenes in images
  - Facial Recognition: Detecting and verifying faces in security, biometrics and authentication
  - Object Detection: Identifying and locating multiple objects in an image or video
  - Medical Imaging: Detecting anomalies in MRIs, X-rays and CT scans
  - Optical Character Recognition (OCR): Extracting text from scanned documents or images
  - Autonomous Navigation: Assisting self-driving cars and robots in real-time scene understanding

- NLP (Natural Language Processing):
    - Speech Recognition: Converting spoken language into text
    - Machine Translation: Translating text across languages
    - Chatbots and Virtual Assistants: AI-powered conversation agents
    - Sentiment Analysis: Understanding emotions or options from text
    - Text Summarization: Generating concise summaries of long articles
    - QA (Question Answering): Extracting precise answers from a text or knowledge base
    - Text Generation: AI-generated articles, summaries and creative writing
- Autonomous Systems and Robotics:
    - Self-driving Vehicles: Cars, trucks and drones using AI for navigation
    - Industrial Robots: AI-driven robotic arms in manufacturing
    - Drones: AI-powered navigation and mapping for drones
- Generative AI and Creativity:
    - AI-generated Art and Images: Create new images using AI
    - Text to Speech and Speech Synthesis: Generate human-like voices
    - Music and Video generation: AI-generated music, videos and animations
    - Code generation and Auto-completion: AI-assisted programming
- Predictive Analytics and Decision Making:
    - Anomaly Detection: Identifying fraud, cybersecurity threats and manufacturing defects
    - Risk Assessment: Evaluating risks in credit scoring, insurance and finance
    - Predictive Maintenance: Forecasting equipment failures in factories and infrastructure
    - Market Forecasting: Predicting stock trends, sales and economic fluctuations